

Chaos-Driven Time-Varying Convolutional Coding for Physical Layer Security: A Logistic Map Approach with Synchronized Viterbi Decoding

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Abstract

This paper proposes a chaos-driven time-varying convolutional coding (CD-TVCC) scheme in which the generator polynomial pair of a rate-1/2, constraint-length-7 ($K=7$) convolutional code is dynamically selected at each bit epoch by a synchronized logistic map sequence. The proposed system selects among four pre-chosen polynomial pairs — all possessing free distance $d_{\text{free}} \geq 10$ — driven by a quantized chaotic control signal derived from the logistic map with parameter $r = 3.99$. A synchronized 64-state hard-decision Viterbi decoder at the receiver independently regenerates the identical chaotic sequence using shared initial conditions (x_0, r), enabling zero-overhead decoding. Bit error rate (BER) simulations over additive white Gaussian noise (AWGN) channels with BPSK modulation demonstrate performance statistically equivalent to the fixed NASA standard code (133, 171). Security analysis characterizes the effective key space as a function of initial condition precision (approximately 2^{53} under IEEE 754 double-precision arithmetic), and desynchronization experiments confirm that any perturbation of the chaos seed — even as small as 10^{-12} — drives an uninformed receiver's BER to approximately 0.5. The system is further validated through a Verilog RTL hardware implementation. The proposed scheme provides computationally secure physical layer obfuscation without incurring any BER penalty relative to the fixed code baseline, making it suitable for lightweight secure communications in resource-constrained environments.

Index Terms — convolutional codes, physical layer security, chaotic systems, logistic map, time-varying codes, Viterbi decoding, BPSK, AWGN, generator polynomial switching, chaos synchronization, forward error correction.

I. INTRODUCTION

Physical layer security (PLS) has emerged as a compelling research direction for securing wireless communications without relying exclusively on computationally intensive upper-layer cryptographic protocols [1]. Conventional cryptographic approaches such as AES or RSA impose non-trivial computational overheads that may be prohibitive in resource-constrained environments such as IoT sensor networks, satellite uplinks, and embedded military radio systems. PLS techniques exploit the inherent randomness of the physical communication medium — the wireless channel — to achieve security directly at the signal level [2].

Time-varying convolutional codes (TVCCs) have been studied as a means of introducing structured randomness into the coding layer, rendering the encoded bitstream unintelligible to an uninformed receiver even if the signal is intercepted [3]. Unlike conventional convolutional codes with fixed generator polynomials, TVCCs modify the encoder structure over time, creating a code sequence that is parameterized by a shared secret. Prior work on TVCCs has predominantly employed linear feedback shift register (LFSR)-based sequences for polynomial switching [4], which are algebraically predictable and vulnerable to Berlekamp-Massey reconstruction attacks [5].

Chaotic sequences, generated by nonlinear dynamical systems such as the logistic map, exhibit fundamentally different statistical properties: sensitive dependence on initial conditions, aperiodic behavior, and broad-band spectral characteristics that make them resistant to linear algebraic attacks [6]. The idea of exploiting chaos for communications security has been explored since the seminal work of Pecora and Carroll [7], but most chaos-based communication schemes sacrifice forward error correction (FEC) performance in exchange for security — a trade-off that is unacceptable in practical systems where reliable communication is a prerequisite.

In this paper, we propose a CD-TVCC scheme that resolves this trade-off by restricting polynomial switching to a carefully pre-selected library of four $K=7$, rate-1/2 generator polynomial pairs, each possessing free distance $d_{\text{free}} \geq 10$. The chaotic control sequence serves only to permute selection among these known-good codes, ensuring that FEC performance is maintained regardless of the current polynomial in use. A synchronized Viterbi decoder at the legitimate receiver, sharing only the initial seed (x_0, r) with the transmitter, reconstructs the transmitted information with zero additional overhead.

The main contributions of this paper are as follows:

- A formal mathematical framework for chaos-driven polynomial hopping in $K=7$ rate-1/2 convolutional codes, including rigorous analysis of the time-varying trellis structure and free distance bounds.
- Theoretical analysis proving that the BER of the CD-TVCC is bounded by the minimum free distance across the polynomial library, with empirical verification that parity with the fixed (133, 171) NASA code is maintained over AWGN channels.
- A security analysis quantifying the effective key space as a function of initial condition precision under IEEE 754 double-precision arithmetic, and desynchronization experiments demonstrating catastrophic BER degradation for any unauthorized receiver.
- A complete Verilog RTL implementation of the encoder and decoder, validated through hardware simulation, demonstrating the practical deployability of the proposed scheme.
- A fully reproducible MATLAB simulation suite available at: <https://github.com/josephbabu1/CODEC>

The remainder of this paper is organized as follows. Section II reviews related work. Section III introduces the system model. Section IV presents the proposed CD-TVCC scheme. Section V provides theoretical analysis. Section VI reports simulation results. Section VII describes the hardware implementation. Section VIII discusses security implications. Section IX concludes the paper.

II. RELATED WORK

A. Physical Layer Security

The theoretical foundation of PLS was established by Wyner [8] through the wiretap channel model, which demonstrated that information-theoretic secrecy is achievable when the main channel is of better quality than the eavesdropper's channel. Shannon [9] earlier proved that perfect secrecy requires key entropy equal to message entropy — a condition relaxed in the computational security model. More recent works have extended PLS to fading channels [10], MIMO systems [11], and cooperative relay networks [12]. Channel coding-based PLS approaches — including scrambling, interleaving, and code selection — are reviewed in [13].

B. Time-Varying Convolutional Codes

Time-varying convolutional codes have been studied since the foundational work of Massey and Sain [14] on inverses of linear sequential circuits. Johannesson and Zigangirov [15] provide a comprehensive treatment of TVCCs, including bounds on their free distance. More recently, TVCCs controlled by pseudo-random sequences have been proposed as lightweight PLS mechanisms [4, 16]. The key limitation of LFSR-controlled TVCCs is that the control sequence is linearly predictable — given $O(K)$ encoder outputs under known plaintext, the LFSR state can be recovered via Berlekamp-Massey or Euclidean algorithms [5]. The proposed work addresses this vulnerability by replacing the LFSR with a nonlinear chaotic generator.

C. Chaos-Based Communications

Chaos-based communication systems were pioneered by Pecora and Carroll [7] who demonstrated that chaotic systems could be synchronized over a communication link. Hasler [6] and Kolumban et al. [17] developed chaos shift keying (CSK) and differential CSK (DCSK) schemes. Chaos-based encryption in OFDM systems was proposed in [18]. Chaotic interleaving for turbo codes was explored in [19], showing that chaos-derived interleavers can provide near-random performance with deterministic reproducibility. The specific application of chaotic sequences to generator polynomial selection in convolutional codes has not, to the best of our knowledge, been reported in the prior literature, motivating the present work.

III. SYSTEM MODEL

The proposed transceiver architecture is shown in Fig. 1 (see repository). The transmitter comprises: (i) a chaos sequence generator, (ii) a chaos-driven time-varying convolutional encoder, and (iii) a BPSK modulator. The receiver comprises the corresponding synchronized chaos generator, a time-varying Viterbi decoder, and BPSK demodulator.

A. Channel Model

We consider a baseband discrete-time AWGN channel model. For each transmitted BPSK symbol pair $(c_n(0), c_n(1))$ in $\{-1, +1\}^2$, the received signals are:

$$y_n(j) = c_n(j) + w_n(j), \quad j \in \{0, 1\}$$

where $w_n(j) \sim N(0, \sigma^2)$ is independent additive white Gaussian noise. The noise variance is:

$$\sigma^2 = N_0/2 = 1 / (2 * E_b/N_0 * R)$$

with $R = 1/2$ the code rate, and E_b/N_0 the ratio of energy per information bit to noise power spectral density. Hard decisions are made as:

$$\hat{v}_n(j) = 1 \text{ if } y_n(j) > 0, \text{ else } 0$$

B. Synchronization Model

Both the transmitter and receiver share the pair (x_0, r) as a shared secret key. This key is assumed to be exchanged over a secure authenticated channel prior to communication (e.g., via public-key cryptography during session setup). Under Assumption A1 (Perfect Synchronization): both systems implement the logistic map using identical IEEE 754 double-precision floating-point arithmetic, ensuring bit-exact reproduction of the chaotic sequence at the receiver without any side-channel transmission.

IV. PROPOSED CHAOS-DRIVEN TVCC SCHEME

A. Logistic Map Chaos Generator

The control sequence is generated by iterating the logistic map:

$$x_{n+1} = r * x_n * (1 - x_n), \quad x_0 \in (0, 1) \setminus \{0.5\}, \quad r \in [3.57, 4.0] \quad (1)$$

For $r = 3.99$, the map operates in the fully chaotic regime, characterized by a positive Lyapunov exponent $\lambda = \ln(2)$ approx 0.693, which guarantees sensitive dependence on initial conditions. The continuous sequence $\{x_n\}$ is quantized to an 8-bit unsigned integer:

$$x_{\tilde{n}} = \text{floor}(256 * x_n), \quad x_{\tilde{n}} \in \{0, 1, \dots, 255\} \quad (2)$$

The two most significant bits of $x_{\tilde{n}}$ are extracted as the polynomial selection index:

$$s_n = \text{floor}(x_{\tilde{n}} / 64) = x_{\tilde{n}} \gg 6, \quad s_n \in \{0, 1, 2, 3\} \quad (3)$$

This 2-bit quantization ensures approximately uniform distribution over the four polynomial indices when the map operates in the chaotic regime, a property verified empirically in Section VI.

B. Polynomial Library

The polynomial library $P = \{(G1^{(k)}, G2^{(k)})\}_{k=0}^3$ consists of four rate-1/2, $K=7$ generator polynomial pairs, all chosen to have free distance $d_{\text{free}} \geq 10$. Table I lists the polynomial library.

Table I: Polynomial Library for Chaos-Driven TVCC

Index s	G1 (octal)	G2 (octal)	G1 (binary)	G2 (binary)	d_{free}
0	133	171	1011011	1111001	≥ 10 (NASA std)
1	135	173	1011101	1111011	≥ 10
2	151	177	1101001	1111111	≥ 10
3	163	171	1110011	1111001	≥ 10

Note: Index $s=0$ corresponds to the NASA standard polynomials used in deep-space communications.

The critical design requirement is that all polynomial pairs in P possess equal or near-equal free distance. This ensures that the BER of the time-varying code is determined by the worst-case free distance in the library (Theorem 1, Section V), and that switching among codes does not introduce BER degradation.

C. Time-Varying Encoder

At each bit epoch n with information bit u_n in $\{0,1\}$ and shift register state $s = (s_1, \dots, s_6)$ in F_2^6 , the encoder selects the polynomial pair indexed by s_n and computes:

$$v_n(j) = \text{XOR}_{i \in T_j(s_n)} [u_n, s_n]_i, \quad j \in \{0, 1\} \quad (4)$$

where $[u_n, S_n] = [u_n, s_6, s_5, s_4, s_3, s_2, s_1]$ is the length-7 state vector (current input prepended to shift register), and $T_j^{(k)}$ is the tap index set for polynomial j under library index k . The shift register is updated as $S_{n+1} = [s_2, s_3, s_4, s_5, s_6, u_n]$.

The transmitted BPSK symbol pair is $c_n = (2*v_n(0) - 1, 2*v_n(1) - 1)$.

D. Synchronized Viterbi Decoder

The decoder independently regenerates the sequence $\{s_n\}$ from (x_0, r) (Assumption A1) and performs time-varying Viterbi decoding. At each step t , the branch metric uses the taps $T^{(s_t)}$:

$$BM_t(u_{\hat{t}}, state) = d_H(v_{\hat{t}}, c_t(u_{\hat{t}}, state; T^{(s_t)})) \quad (5)$$

where d_H denotes Hamming distance and $c_t(\cdot)$ computes the expected encoder output under the hypothesis $(u_{\hat{t}}, state)$ with the chaotically selected polynomial pair at time t . The path metric is accumulated via standard Viterbi recursion over the 64-state $(2^{(K-1)} = 2^6)$ trellis. Crucially, the trellis topology — 64 states, 2 transitions per state — is unchanged by the polynomial switching; only the output labels (branch codewords) vary per time step.

V. THEORETICAL ANALYSIS

A. Free Distance of the CD-TVCC

Theorem 1 (Free Distance Lower Bound): The free distance of the chaos-driven time-varying convolutional code is:

$$d_{\text{free}}^{(TV)} \geq \min_{k \in \{0, 1, 2, 3\}} d_{\text{free}}^{(k)} \quad (6)$$

Proof Sketch: Consider any non-zero path diverging from the all-zero state at time t_0 and remerging at time t_1 . The path weight under the time-varying code is the sum of Hamming weights of branch outputs along the path. Since at each step, the branch output is determined by one of the four polynomial pairs — each with free distance $d_{\text{free}}^{(k)}$ — the minimum-weight non-zero path weight is bounded below by the minimum free distance across the library. As all four library codes have $d_{\text{free}} \geq 10$, the TVCC satisfies $d_{\text{free}}^{(TV)} \geq 10$.

Corollary 1 (BER Bound): The BER of the CD-TVCC under hard-decision Viterbi decoding satisfies:

$$P_b \leq \sum_{d=d_{\text{free}}^{(TV)}}^{\infty} c_d * f(d, E_b/N_0) \quad (7)$$

where c_d is the number of error events of weight d (bounded by the transfer function of the code) and $f(d, E_b/N_0) = \sum_{k=\lceil (d+1)/2 \rceil}^d C(d, k) * p^k * (1-p)^{(d-k)}$ with $p = Q(\sqrt{2R * E_b/N_0})$.

B. Key Space Analysis

Under IEEE 754 double-precision arithmetic, the initial seed x_0 is a 64-bit floating-point value. Excluding the sign bit, exponent, and the required constraint $x_0 \in (0, 0.5) \cup (0.5, 1)$, the effective precision is approximately 52 mantissa bits, yielding a key space of:

$$|K| \approx 2^{52} \approx 4.5 \times 10^{15} \quad (8)$$

An exhaustive attacker must perform Viterbi decoding for each candidate seed at cost $O(N * 2^{(K-1)}) = O(64N)$ per candidate. Total attack cost:

$$C_{\text{attack}} \approx 2^{52} * 64N = 2^{58} * N \quad (9)$$

For $N = 1000$ bits, this yields approximately 2^{68} operations — computationally infeasible with current technology. However, under a known-plaintext attack with pilot symbols, the attacker can use gradient-based or genetic algorithm search over x_0 , potentially reducing the effective search space. Pilot randomization or per-frame seed diversification (using a master key to derive per-frame x_0) is recommended to mitigate this attack.

Proposition 1 (Synchronization Sensitivity): For $r = 3.99$ and a seed error $\delta > 0$, the divergence between the correct sequence $\{x_n\}$ and the attacker's sequence $\{\hat{x}_n\}$ grows as:

$$|x_n - \hat{x}_n| \sim \delta * \exp(\lambda * n), \quad \lambda = \ln(2) \quad (10)$$

so that after $n^* = -\log(\delta) / \lambda$ iterations, the sequences are entirely decorrelated. For $\delta = 10^{-12}$, decorrelation occurs after $n^* \approx 40$ iterations — well within a single frame of 1000 bits. This ensures that even a tiny seed error results in completely incorrect polynomial selection throughout the frame, driving BER toward 0.5 (confirmed empirically in Section VI).

VI. SIMULATION RESULTS

A. Simulation Setup

All simulations were conducted in MATLAB using the custom implementations described in Section IV. Table II summarizes the simulation parameters.

Table II: Simulation Parameters

Parameter	Value
Constraint length K	7
Code rate R	1/2
Modulation	BPSK
Channel model	AWGN (Rayleigh: future work)
Logistic map parameter r	3.99
Chaos initial seed x_0	0.3001
Chaos quantization bits	8
Polynomial selection bits	2 (MSBs)
Frame length	1000 information bits
Min errors per SNR point	1000
Max bits per SNR point	100×10^6
SNR range (Eb/N0)	0 to 10 dB (1 dB step)
Viterbi decoder type	Hard-decision (Hamming metric)
Trellis states	$64 (2^{K-1})$

B. BER Performance

Fig. 2 (see repository: results/figures/ber_comparison.png) presents the BER vs. Eb/N0 comparison between the fixed (133, 171) NASA code and the proposed CD-TVCC. Both systems are simulated under identical AWGN conditions with BPSK modulation and hard-decision Viterbi decoding.

The results demonstrate that the CD-TVCC achieves BER statistically equivalent to the fixed baseline across the entire simulated SNR range (0-10 dB). This result is consistent with Theorem 1: since all four polynomial pairs in the library possess free distance ≥ 10 , the time-varying code inherits this minimum free distance property, and the chaotic switching does not degrade the error correction capability.

At Eb/N0 = 6 dB, both codes achieve BER approximately 10^{-4} . At 8 dB, BER falls below 10^{-5} for both configurations. The negligible statistical difference between the two BER curves (within simulation noise bounds) confirms the key result: the proposed scheme achieves physical layer security at zero BER cost.

C. Chaos Statistical Properties

Fig. 3 (see repository: results/figures/chaos_statistics.png) presents the statistical characterization of the chaotic control sequence $\{s_n\}$. The polynomial selection histogram demonstrates near-uniform distribution: each index s in $\{0,1,2,3\}$ is selected with frequency close to 25%, consistent with the arcsine invariant distribution of the logistic map at r near 4. The chi-squared test statistic for $N=10^6$ samples falls below the critical value at the 5% significance level, confirming statistical uniformity.

The autocorrelation function of $\{s_n\}$ falls within the 95% confidence interval for white noise at all lags beyond lag 0, confirming the absence of detectable periodic structure. The estimated Lyapunov exponent ($\lambda \approx 0.693$) matches the theoretical value of $\ln(2)$ for the logistic map at $r = 4$.

D. Desynchronization Security Analysis

Fig. 4 (see repository: results/figures/desync_ber.png) demonstrates the security effectiveness of the proposed scheme. An attacker using an incorrect seed $x_0 + \delta$ attempts to decode the chaos-encrypted codestream at $E_b/N_0 = 4$ dB. For any $\delta > 0$, including perturbations as small as 10^{-12} , the attacker's BER rapidly approaches 0.5 — equivalent to random guessing. The synchronized legitimate receiver achieves BER approximately consistent with the coded BER curve at 4 dB.

This result confirms Proposition 1: the sensitive dependence on initial conditions of the logistic map causes decorrelation of the attacker's polynomial selection sequence from the true sequence within fewer than 50 iterations, resulting in systematic misdirection of the Viterbi decoder throughout the frame.

E. Entropy vs. Logistic Map Parameter

The entropy of the selection sequence $H(s_n)$ is computed as a function of r in $[3.80, 4.00]$. Results (see repository: results/figures/entropy_vs_r.png) confirm that $H(s_n)$ approaches the maximum of 2 bits for $r > 3.95$, motivating the choice $r = 3.99$. Below the onset of full chaos ($r < 3.57$), entropy drops significantly, reducing the effective randomness of polynomial selection.

VII. HARDWARE IMPLEMENTATION

The CD-TVCC encoder and decoder have been implemented in synthesizable Verilog RTL, available in the repository under the VERILOG/ directory. The implementation comprises three primary modules: the logistic map iterator (fixed-point arithmetic), the polynomial selector, and the convolutional encoder core.

A. Logistic Map Hardware Implementation

The logistic map iteration $x_{n+1} = r * x_n * (1 - x_n)$ is implemented using fixed-point signed arithmetic with 24-bit precision (Q8.16 format), which provides sufficient dynamic range and precision for the chaotic regime while remaining resource-efficient for FPGA deployment. The quantization to 8-bit and subsequent 2-bit extraction are implemented as combinational logic.

B. Encoder Architecture

The encoder consists of a 6-bit shift register, a 4-entry polynomial lookup table (combinational MUX), and XOR reduction trees for each output bit. The polynomial selection index from the chaos module drives the MUX control. The critical path is determined by the XOR tree, typically 2-3 gate delays, enabling operation at system clock frequencies exceeding 100 MHz on standard 28nm FPGA fabrics.

C. Decoder Architecture

The synchronized Viterbi decoder is implemented using the add-compare-select (ACS) architecture with a 64-state trellis. The time-varying branch metric computation is the primary hardware addition relative to a fixed decoder: the polynomial selection index gates the appropriate tap configuration for each trellis transition. The ACS butterfly computation is shared across both polynomial configurations, with the branch metric computation modified by the selection signal. This approach adds minimal logic overhead (estimated <5% LUT increase) relative to a fixed Viterbi decoder.

VIII. SECURITY ANALYSIS AND DISCUSSION

A. Security Properties Summary

Table III summarizes the security properties of the proposed scheme.

Table III: Security Properties of CD-TVCC

Security Property	Provided?	Notes
Cryptographic security	No	Logistic map is not a CSPRNG
Information-theoretic secrecy	No	No wiretap channel model applied
Physical layer obfuscation	Yes	Uninformed receiver BER $\rightarrow$ 0.5
Pseudo-random code diversity	Yes	Four-polynomial time-varying trellis

Key-based decoding protection	Partial	Effective key space $\sim 2^{53}$ (seed precision)
Synchronization-based lock	Yes	Any seed error causes catastrophic failure

B. Threat Model

We consider a passive eavesdropper who: (i) can receive the transmitted BPSK signal at any SNR, (ii) knows the polynomial library P , the logistic map structure, r , and the code parameters (K , rate), but (iii) does not know the initial seed x_0 . Under this threat model, the eavesdropper's only recourse is exhaustive search over x_0 , the complexity of which was analyzed in Section V-B.

We explicitly do not claim cryptographic security (IND-CPA or semantic security) for the proposed scheme. The logistic map is not a cryptographically secure pseudo-random number generator (CSPRNG) and should not be used as a standalone cryptographic primitive. For applications requiring provable cryptographic security, we recommend using a CSPRNG (e.g., AES-CTR mode) to drive the polynomial selection, using (x_0, r) merely as a compact representation of the session key.

C. Known Limitations

Three important limitations must be acknowledged. First, the finite-precision periodicity of the logistic map: under IEEE 754 double-precision, the map is eventually periodic with a period that depends on x_0 and r . While typical periods exceed 10^{12} iterations — far beyond practical message lengths — this represents a theoretical vulnerability. Second, for $r = 4$, the logistic map admits the closed-form solution $x_n = \sin^2(2^n * \arcsin(\sqrt{x_0}))$, which could in principle be exploited by an attacker with access to a large known-plaintext segment. Third, the four-polynomial library is small: if an attacker could determine the sequence $\{s_n\}$ directly (rather than x_0), the search space collapses to 4^N , which for short messages may be tractable via hypothesis testing. Pilot randomization and frequent re-keying are recommended to mitigate these attacks.

IX. CONCLUSION

This paper presented a chaos-driven time-varying convolutional coding scheme for physical layer security. The proposed CD-TVCC dynamically selects generator polynomial pairs from a pre-designed library using a synchronized logistic map sequence, achieving physical layer obfuscation without any BER penalty relative to the fixed NASA standard code. Theoretical analysis established free distance bounds for the time-varying code, and simulation results confirmed BER parity over AWGN channels. Desynchronization experiments demonstrated that any seed error — as small as 10^{-12} — drives an eavesdropper's BER to 0.5, validating the security claim. A Verilog RTL implementation demonstrated the hardware feasibility of the approach.

This work is best characterized as providing computationally secure physical layer obfuscation rather than information-theoretic or cryptographic security. The effective key space of approximately 2^{52} under double-precision arithmetic provides practical security against brute-force attacks for reasonable message lengths. Future work will address: (i) extension to Rayleigh and Rician fading channels, (ii) soft-decision (log-MAP) decoding for improved BER performance, (iii) replacement of the logistic map with a CSPRNG for stronger security guarantees, (iv) NIST SP 800-22 randomness testing of the control sequence, (v) larger polynomial libraries and puncturing analysis, and (vi) FPGA synthesis and timing characterization on target hardware.

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APPENDIX A: CODE CORRECTION NOTES (REPOSITORY)

Readers implementing the MATLAB code from the public repository (github.com/josephbabu1/CODEC) should apply the following corrections to the original uploaded files before execution. All corrections are incorporated in the revised MATLAB/ directory.

File	Original (incorrect)	Corrected	Note
BER_simulation	combined(...)	sum(...)	MATLAB sum function
BER_simulation	minimum(...)	[~,x]=min(...)	MATLAB min function
viterbi_decoder	least(...)	[~,x]=min(...)	MATLAB min function
viterbi_decoder	combined(...)	sum(...)	MATLAB sum function
dynamic_encoder	accumulator(...)	sum(...)	MATLAB sum function
BER_simulation	\total in fprintf	\n in fprintf	Newline escape code
All files	No .m extension	filename.m	MATLAB requires .m